

Adrian Röllin

Curriculum Vitae

Education

- 11/2002–11/2006 **Ph.D. in Mathematics**, *University of Zürich, Switzerland*, under supervision of A. D. Barbour.
- 10/1996–10/2002 **M.Sc. in Mathematics**, *University of Zürich, Switzerland*, minor subjects: computer science, actuarial science.

Positions

- since 07/2021 Professor, Department of Statistics and Data Science, NUS
- 01/2015–06/2021 Associate Professor, Department of Statistics and Applied Probability, NUS
- 07/2009–12/2014 Assistant Professor, Department of Statistics and Applied Probability, NUS
- 09/2008–06/2009 Postdoc, Department of Mathematics, NUS
- 05/2008–07/2008 Postdoc, Institute of Social and Preventive Medicine, University of Bern
- 04/2007–03/2008 Postdoc, Department of Statistics, University of Oxford

Other Appointments

- since 07/2018 Honorary appointment, Saw Swee Hock School of Public Health, NUS
- since 07/2016 Honorary appointment, Department of Mathematics, NUS

Awards and Recognitions

- 2019 IMS Fellow (Institute of Mathematical Statistics)
- 2019 ISI Elected Member (International Statistical Institute)
- 2019 Dean's Chair, Faculty of Science, NUS
- 2010 Young Scientist Award, Faculty of Science NUS
- 2007 PhD thesis awarded for scientific excellence from the Faculty of Science, University of Zürich

Editorial Work

- since 01/2021 Associate Editor, *Bernoulli Journal*
- since 10/2020 Associate Editor, Applied Probability Trust (*Advances in Applied Probability* and *Journal of Applied Probability*)
- 2013–2021 Associate Editor, *Probability in the Engineering and Informational Sciences*

Professional Committee Work

- since 2025 Vice President, Singapore Mathematical Society
- 2023–2027 Member of the Board of Directors of IASC-ARS (International Association for Statistical Computing, Asian Regional Section)
- 2024–2025 Committee on Nominations, Institute of Mathematical Statistics
- 2020–2025 Ethel Newbold Prize Committee, Bernoulli Society (Chair 2024–2025)
- 2020–2025 Management Committee, Singapore Mathematical Society

Service appointments

- 2021–2024 Head, Department of Statistics and Data Science, NUS
- 2018–2021 Deputy Director, Institute for Mathematical Sciences, NUS
- 2018–2021 Deputy Head, Department of Statistics and Applied Probability, NUS
- 2018–2021 Faculty Promotion and Tenure Committee, Faculty of Science, NUS

Memberships in Professional Organisations

- American Mathematical Society
- Bernoulli Society
- Institute of Mathematical Statistics
- Intern. Assoc. for Statistical Computing
- International Statistical Institute
- Singapore Mathematical Society

Research Interests

- Distributional approximations in probability and statistics via Stein's method
- Random graphs and other random combinatorial objects
- Epidemiological modelling
- Applied statistical modelling and analysis in life sciences
- Statistical applications of deep neural networks

Research Group

Postdocs

- 2023–2025 Francesca Cottini, joint Luxembourg-Singapore research grant, NRF Luxembourg
- 2022–2023 Yuanfei Huang, under MOE Tier 2 research grant
- 2022–2023 Tianshu Cong, under MOE Tier 2 research grant
- 2020–2022 Van Hao Can, under MOE Tier 2 research grant
- 2019–2021 Zhuosong Zhang, under MOE Tier 2 research grant
- 2018–2021 Gursharn Kaur, under Faculty of Science RSB program
- 2019–2020 Arturo Jaramillo Gil, joint with *University of Luxembourg*, under Luxembourg NRF
- 2019–2020 Xiaochuan Yang, joint with *University of Luxembourg*, under Luxembourg NRF

Ph.D. students

- 2021–2026 Ryo Imai, *Stein's Method for Multivariate Translated Poisson Approximation*
- 2018–2023 Shang Li, *Stein's Method for Total Variation Approximation With Applications to Analysis of Computer Algorithms.*
- 2016–2020 Wai Hoh Tang, *Deep Learning in Fluorescence Spectroscopy.*
- 2010–2014 Daniel Paulin (co-supervised with Louis H. Y. Chen), *Stein's method for Concentration of Measure.*

Meetings

Upcoming events

- 08/2025 Invited speaker, *Stein's Method in Stochastic Geometry, Statistical Learning, and Optimisation*, Mathematisches Forschungsinstitut Oberwolfach, Germany

Past events

- 07/2025 *44th Conference on Stochastic Processes and their Applications*, Wrocław, Poland
- 05/2025 Invited speaker, Berliner Kolloquium, Freie Universität Berlin, Germany
- 05/2025 Invited speaker, Probability Seminar, WIAS, Technische Universität Berlin, Germany
- 12/2024 Invited speaker, *Mathematical Foundations of Network Models and Their Applications*, BIRS/CMI Workshop, Chennai, India
- 09/2024 Invited speaker, Probability Seminar, ICTS-TIFR, Bangalore, India
- 09/2024 Invited speaker, Probability Seminar, University of Melbourne, Australia

- 08/2024 Invited speaker, *Bernoulli–IMS World Congress*, Bochum, Germany
- 01/2024 Invited participant, *Discussion Meeting on Probability* at ICTS-TIFR Bangalore, India
- 01/2024 Invited speaker, *IMS Pacific RIM*, Melbourne, Australia
- 08/2023 Invited speaker, *Permutation and Causal Inference Connections and Applications*, IMSI Chicago, United States
- 03/2023 Invited speaker, *UK Easter Probability Meeting*, Manchester, United Kingdom
- 09/2022 Invited speaker, *Graph Limits, Non-Parametric Models, and Estimation*, Simons Institute, UC Berkeley
- 04/2022 Invited speaker, *Advances in Stein's Method and its Applications in Machine Learning and Optimization*, BIRS Workshop, Banff, Canada
- 03/2022 Invited speaker (hybrid), *Population Dynamics and Statistical Physics in Synergy*, Oberwolfach, Germany
- 10/2020 Invited speaker (online talk), *Stochastics Seminar*, Georgia Tech, USA
- 07/2020 Invited speaker (online talk), *Probability Victoria Seminar*, Australia
- 2020 Invited participant, *Discussion Meeting on Stochastic Analysis, Geometry, and Random Fields* at ISI Bangalore, India
- 2019 Invited to conduct 4-day mini-lecture series as part of graduate course *Introduction to Causal Inference and Stein's method*, Universities of Mannheim and Heidelberg, Germany
- 2019 Invited-session organiser, *ISI World Statistics Congress*, Kuala Lumpur, Malaysia
- 2019 Invited speaker, *Mathematical Colloquium*, Institute for Mathematics, University of Osnabrück, Germany
- 2018 Invited speaker, *Workshop on High-Dimensional Phenomena in Probability*, Bochum, Germany
- 2018 Invited speaker, *Workshop in Honour of Carl-Gustav Esseen's 100th Birthday*, Uppsala, Sweden
- 2018 Invited participant, *AIM Workshop on Stein's Method and Applications in High-Dimensional Statistics*, San Jose, United States
- 2018 Invited-session speaker, *IMS Asia Pacific Rim Meeting*, Singapore
- 2018 Invited-session speaker, *9th International Workshop on Applied Probability*, Budapest, Hungary
- 2018 Invited speaker, *International Conference in Statistics and Probability in Memory of 125 P. C. Mahalanobis*, Kolkata, India
- 2017 Invited speaker at *Melbourne-Singapore Probability and Statistics Forum*, Melbourne, Australia
- 2017 Invited speaker, *Probability Meeting at ISI Bangalore*, Bangalore, India
- 2016 Invited-session organiser and invited-session speaker, *IMS Asia Pacific Rim Meeting*, Hong Kong
- 2016 Invited speaker, *Asian Mathematical Conference*, Bali, Indonesia
- 2015 Invited-session speaker, *Stochastic Processes and Applications*, Oxford, United Kingdom
- 2014 Invited-session speaker, *IMS Annual Meeting*, Sydney, Australia
- 2014 Invited speaker, *International Colloquium on Stein's Method, Concentration Inequalities, and Malliavin Calculus*, Nantes, France
- 2013 Invited speaker for lecture series at the University of Melbourne, Melbourne, Australia
- 2013 Invited speaker, *Workshop on New Directions in Probability*, ISI Bangalore, India
- 2013 Invited-session speaker, *IMS–China International Conference on Statistics and Probability*, Chengdu, China
- 2013 Invited-session speaker, *Statistics, Science, and Society: New Challenges and Opportunities*, international conference organised by the International Indian Statistical Association, Chennai, India

- 2012 Invited speaker of contributed session, *8th World Congress in Probability and Statistics*, Istanbul, Turkey
- 2012 Invited-session speaker, *International Workshop in Applied Probability*, Jerusalem, Israel
- 2008 Invited speaker, *IMS Workshop on Stein's method*, Singapore, 2008.
- 2007 *32nd conference on Stochastic Processes and their Applications*, Urbana-Champaign, USA, 2007.
- 2007 Invited speaker, *2nd International Workshop on recent advances in probability, statistics & financial stochastics*, Middlesex University Business School, UK, 2007.
- 2006 *XXVI European Meeting of Statisticians*, Torun, Poland, 2006.
- 2003 *IMS Programme on Stein's method*, Singapore, 2003.

Conference Organiser

- 07/2028 Chair of Local Organising Committee, *Bernoulli-IMS World Congress in Probability and Statistics*, Singapore
- 2027 Co-Chair, *One Hundred Years (or so) of Epidemiological Modelling*, Institute for Mathematical Sciences, NUS
- 11/2026 Co-Chair, *FIM-IMS Joint Workshop on Mathematics of Data Science*, Institute for Mathematical Sciences, Singapore
- 11/2026 Co-Chair, *IMS Young Mathematical Scientists Forum — Statistics and Data Science*, Institute for Mathematical Sciences, Singapore
- 05/2026 Co-Chair, *SASI — Singapore–Abu Dhabi–Shanghai–India Probability Meeting*, NYU Shanghai, China
- 12/2025 Member of the Scientific Programme Committee, *IASC-ARS Meeting 2025*, Ho Chi Minh City, Vietnam
- 11/2025 Co-Chair, *IMS Young Mathematical Scientists Forum — Statistics and Data Science*, Institute for Mathematical Sciences, Singapore
- 10/2025 Chair of Scientific Programme Committee, *65th ISI World Statistics Congress*, The Hague, Netherlands
- 11/2023 Co-Chair, *IMS Young Mathematical Scientists Forum — Statistics and Data Science*, NUS, Singapore
- 2022 *Stein's Method — The Golden Anniversary*, Institute for Mathematical Sciences, NUS
- 2022 *A Tale of Rare Events — Symposium in Honour of Louis Chen on His 80th Birthday*, NUS
- 2022 *SASI — Singapore–Abu Dhabi–Shanghai–India Probability Meeting*, NYU Abu Dhabi Institute, Abu Dhabi
- 2021 *SASI — Singapore–Abu Dhabi–Shanghai–India Probability Meeting*, NYU Abu Dhabi
- 2019 Co-Chair, *SASI — Singapore–Abu Dhabi–Shanghai–India Probability Meeting*, Institute for Mathematical Sciences, Singapore
- 2019 Co-Chair, *Symposium in Memory of Charles Stein [1920 - 2016]*, Institute for Mathematical Sciences, NUS
- 2016 Co-Chair, *Second Melbourne–Singapore Probability and Statistics Forum*, Institute for Mathematical Sciences, Singapore
- 2015 *Workshop on New Directions in Stein's Method*, Institute for Mathematical Sciences, Singapore
- 2010 *Workshop on Epidemiology of Infectious Diseases: Emerging Challenges*, Institute for Mathematical Sciences, NUS

Publications

Preprints

- [52] J. Fulman and A. Röllin (2026). Stein's method and the modular behavior of Eulerian numbers. [arXiv:2603.21464](https://arxiv.org/abs/2603.21464)

- [51] S. Athreya, F. Hollander, A. Röllin (2025). Co-evolving vertex and edge dynamics in dense graphs. *arXiv:2504.06493*
- [50] Y. Huang, A. Röllin (2024). The SIR epidemic on a dynamic Erdős-Rényi random graph. *arxiv.org:2404.12566*
- [49] V. H. Can, A. Röllin. Mean-field spin models – Fluctuation of the magnetization and maximum likelihood estimator. *arXiv:2312.07313*.
- [48] L. H. Y. Chen and A. Röllin. Stein couplings for normal approximation. *arXiv:1003.6039*.

Journal articles

- [47] S. Athreya, F. den Hollander and A. Röllin (2025). The Moran model with random resampling rates. *Ann. Appl. Probab.* **35**, 1852–1868
- [46] W. H. Tang, S. R. Sim, D. Y. K. Aik, A. V. S. Nelanuthala, T. Athilingam, A. Röllin, T. Wohland (2024). Deep learning reduces data requirements and allows real-time measurements in Imaging FCS. *Biophysical Journal* **123**, 1–12.
- [45] A. Röllin and Z.-S. Zhang (2023). Dense multigraphon-valued stochastic processes and edge-changing dynamics in the configuration model. *Ann. Appl. Probab.* **33**, 3207–3239
- [44] A. Röllin (2022). Kolmogorov bounds for the normal approximation of the number of triangles in the Erdős-Rényi random graph. *Probab. Engrg. Inform. Sci.* **36**, 747–773.
- [43] J. Fulman and A. Röllin (2022). Stein’s method, heat kernel, and linear functions on the orthogonal groups. *J. Algebra* **607**, 272–285.
- [42] M. A. Mendieta-Serrano, S. Dhar, B. H. Ng, R. Narayanan, J. J. Y. Lee, H. T. Ong, P. J. Y. Toh, A. Röllin, S. Roy, and T. E. Saunders (2022). Slow muscles guide fast myocyte fusion to ensure robust myotome formation despite the high spatiotemporal stochasticity of fusion events. *Dev. Cell* **57**, 2095–2110.e5.
- [41] L. H. Y. Chen, A. Röllin, A. Xia (2021). Palm theory, random measures and Stein couplings. *Ann. Appl. Probab.* **31**, 2881–2923.
- [40] G. Kaur and A. Röllin (2021). Higher-order fluctuations in dense random graph models. *Electronic J. Probab.* **26**, article no. 139, 1–36.
- [39] X. Fang, H. L. Gan, S. Holmes, H. Huang, E. A. Peköz, A. Röllin, and W. Tang (2021). Arcsine laws for random walks generated from random permutations with applications to genomics. *J. Appl. Probab.* **58**, 851–867.
- [38] S. Athreya, F. den Hollander and A. Röllin (2021). Graphon-valued stochastic processes from population genetics. *Ann. Appl. Probab.* **31**, 1724–1745.
- [37] W. H. Tang and A. Röllin (2021). Model identification for ARMA time series through convolutional neural networks. *Decis. Support Syst.* **146**, 113544.
- [36] J. Sankaran, H. Balasubramanian, W. H. Tang, X. W. Ng, A. Röllin, and T. Wohland (2021). Simultaneous spatiotemporal super-resolution and multi-parametric fluorescence microscopy. *Nat. Commun.* **12**, no. 1748.
- [35] L. H. Y. Chen, L. Goldstein and A. Röllin. Stein’s method via induction (2020). *Electronic J. Probab.* **25**, article no. 132, 49 pp.
- [34] J. Fulman and A. Röllin (2020). Stein’s method and Narayana numbers. *Statist. Probab. Lett.* **165**, article no. 108835, 1–9.
- [33] E. A. Peköz, N. Ross and A. Röllin (2020). Exponential and Laplace approximation for occupation statistics of branching random walk. *Electronic J. Probab.* **25**, article no. 55, 1–22.
- [32] A. K. J. Teo, K. Prem, M. I. C. Chen, A. Röllin, M. L. Wong, H. H. La and A. R. Cook (2019). Estimating the size of key populations for HIV in Singapore using the network scale-up method. *Sex. Transm. Infect.* **95**, 602–607.
- [31] A. D. Barbour, A. Röllin, and N. Ross (2019). Error bounds in local limit theorems using Stein’s method. *Bernoulli* **25**, 1076–1104.
- [30] A. D. Barbour and A. Röllin (2019). Central limit theorems in the configuration model. *Ann. Appl. Probab.* **29**, 1046–1069.

- [29] E. A. Peköz, A. Röllin and N. Ross (2019). Pólya urns with immigration at random times. *Bernoulli* **25**, 189–220.
- [28] A. D. Barbour and A. Röllin (2018). A central limit theorem for the gossip process. *Electron. J. Probab.* **23**, article no. 123, 1–37.
- [27] A. Röllin (2018). On quantitative bounds in the mean martingale central limit theorem. *Statist. Probab. Lett.* **138**, 171–176
- [26] S. Athreya and A. Röllin (2018). Respondent driven sampling and sparse graph convergence. *Electron. Comm. Probab.* **23**, article no. 3, 1–12.
- [25] H. L. Gan, A. Röllin, N. Ross. Dirichlet approximation of equilibrium distributions in Cannings models with mutation. *Adv. Appl. Probab.* **49**, 927–959.
- [24] E. A. Peköz, A. Röllin and N. Ross (2017). Joint degree distributions of preferential attachment random graphs. *Adv. Appl. Probab.* **49**, 368–387.
- [23] S. Athreya and A. Röllin (2016). Dense graph limits under respondent-driven sampling. *Ann. Appl. Probab.* **26**, 2193–2210.
- [22] E. A. Peköz, A. Röllin and N. Ross (2016). Generalized gamma approximation with rates for urns, walks and trees. *Ann. Probab.* **44**, 1776–1816.
- [21] X. Fang and A. Röllin (2015). Rates of convergence for multivariate normal approximation with applications to dense graphs and doubly indexed permutation statistics. *Bernoulli* **21**, 2157–2189.
- [20] A. Röllin and N. Ross (2015). Local limit theorems via Landau-Kolmogorov inequalities. *Bernoulli* **21**, 851–880.
- [19] L. H. Y. Chen and A. Röllin (2013). Approximating dependent rare events. *Bernoulli* **19**, 1243–1267.
- [18] K. H. X. Tan, L. Simonella, H. L. Wee, A. Röllin, Y.-W. Lim, W.-Y. Lim, K. S. Chia, M. Hartman, and A. R. Cook (2013). Quantifying the natural history of breast cancer. *Br. J. Cancer* **109**, 2035–2043.
- [17] E. A. Peköz, A. Röllin, and N. Ross (2013). Total variation error bounds for geometric approximation. *Bernoulli* **19**, 610–632.
- [16] A. Röllin (2013). Stein’s method in high dimensions with applications. *Ann. Inst. Henri Poincaré Probab. Stat.* **49**, 529–549.
- [15] E. A. Peköz, A. Röllin and N. Ross (2013). Degree asymptotics with rates for preferential attachment random graphs. *Ann. Appl. Probab.* **23**, 1188–1218.
- [14] A. Röllin (2012). On the optimality of Stein factors. *Probability Approximations and Beyond, Lecture Notes in Statistics 205*, Springer.
- [13] C. L. Althaus, J. C. M. Heijne, S. A. Herzog, A. Röllin and N. Low (2012). Individual and population level effects of partner notification for *Chlamydia trachomatis*. *PLoS ONE* **7**.
- [12] E. A. Peköz and A. Röllin (2011). Exponential approximation for the nearly critical Galton-Watson process and occupation times of Markov chains. *Electron. J. Probab.* **16**, 1381–1393.
- [11] S. Chatterjee, J. Fulman and A. Röllin (2011). Exponential approximation by Stein’s method and spectral graph theory. *ALEA Lat. Am. J. Probab. Math. Stat.* **8**, 197–223.
- [10] E. A. Peköz, A. Röllin (2011). New rates for exponential approximation and the theorems of Rényi and Yaglom. *Ann. Probab.* **39**, 587–608.
- [9] C. L. Althaus, J. C. M. Heijne, A. Röllin and N. Low (2010). Transmission dynamics of *Chlamydia trachomatis* affect the impact of screening programmes. *Epidemics* **2**, 123–131.
- [8] G. Reinert and A. Röllin (2010). Random subgraph counts and U-statistics: multivariate normal approximation via exchangeable pairs and embedding. *J. Appl. Probab.* **47**, 378–393.
- [7] S. Tueckmantel, A. Röllin, A. E. Müller and C. Soligo (2009). Facial correlates of frontal bone pneumatization in strepsirrhine primates. *Mammalian Biology* **74**, 25–35.
- [6] E. A. Peköz, A. Röllin, V. Čekanavičius and M. Shwartz (2009). A three-parameter binomial approximation. *J. Appl. Probab.* **46**, 1073–1085.

- [5] G. Reinert and A. Röllin (2009). Multivariate normal approximation with Stein's method of exchangeable pairs under a general linearity condition. *Ann. Probab.* **37**, 2150–2173.
- [4] A. Röllin (2008). A note on the exchangeability condition in Stein's method. *Statist. Probab. Lett.* **78**, 1800–1806.
- [3] A. Röllin (2008). Symmetric and centered binomial approximation of sums of locally dependent random variables. *Electron. J. Probab.* **13**, 756–776.
- [2] A. Röllin (2007). Translated Poisson approximation using exchangeable pair couplings. *Ann. Appl. Probab.* **17**, 1596–1614.
- [1] A. Röllin (2005). Approximation of sums of conditionally independent variables by the translated Poisson distribution. *Bernoulli* **11**, 1115–1128.

Posters and method articles

- W. H. Tang, S. R. Sim, D. Y. Aik, A. V. S. Nelanthala, T. Athilingam, A. Röllin, T. Wohland (2024). Deep learning reduces data requirements and allows real-time measurements in Imaging FCS. *Biophysical Journal* **123**, 655–666.
- T. Wohland, W. H. Tang, S. R. Sim, D. Y. Aik, A. V. S. Nelanthala, and A. Röllin (2023). FCSNet and ImFCSNet, convolutional neural network approaches to FCS data analysis. *Biophysical Journal* **122**, 429a.
- T. Wohland, W. H. Tang, S. R. Sim, D. Y. Aik, and A. Röllin (2022). Deep learning approaches for imaging fluorescence correlation spectroscopy parameter estimation with limited data sets. *Biophysical Journal* **121**, 533a.
- H. Balasubramanian, J. Sankaran, W. H. Tang, X. W. Ng, A. Röllin, and T. Wohland (2021). SRRF'N'Tirf-FCS: New Insights into EGFR-Cytoskeleton Interactions. *Biophysical Journal* **120**, 186a.
- J. Sankaran, H. Balasubramanian, W. H. Tang, X. W. Ng, A. Röllin, T. Wohland (2021). Preparation of live cell samples for fluorescence spectroscopy and computational super-resolution imaging, 29 March 2021, PROTOCOL (Version 1). Available at Protocol Exchange.
- J. Sankaran, H. Balasubramanian, W. H. Tang, X. W. Ng, A. Röllin, T. Wohland (2021). Simultaneous spatiotemporal computational super-resolution and multi-parametric fluorescence microscopy, 29 March 2021, PROTOCOL (Version 1). Available at Protocol Exchange.

Teaching

- since 07/2009 Lecturing at the National University of Singapore
- 09/2008–06/2009 Teaching assistant at the National University of Singapore
- 04/2007–03/2008 Teaching assistant, supervision of MSc students and admission interviewing at the University of Oxford
- 11/2002–03/2007 Teaching assistant for lectures in mathematics, probability, statistics and bioinformatics, graduate and undergraduate level

Courses taught

Acad. Year	Sem.	Module Code	Module Title
2025/2026	2	ST2334	Probability and Statistics
2025/2026	1	ST5225	Statistical Analysis of Networks
2024/2025	1/2	ST5290	Data Science Industry Project
2024/2025	1	ST5225	Statistical Analysis of Networks
2019/2020	1	ST5214	Advanced Probability Theory
2018/2019	1	ST5214	Advanced Probability Theory
2017/2018	1	ST1131	Introduction to Statistics
2015/2016	2	ST5225	Statistical Analysis of Networks
2015/2016	1	ST5214	Advanced Probability Theory
2014/2015	2	MA4251/ST4238	Stochastic Processes II
2014/2015	1	ST5214	Advanced Probability Theory
2013/2014	2	GEM2900	Understanding Uncert. and Stat. Thinking
2012/2013	2	MA2216/ST2131	Probability
2012/2013	1	ST5214	Advanced Probability Theory
2011/2012	2	MA4251/ST4238	Stochastic Processes II
2011/2012	1	MA3238/ST3236	Stochastic Processes I
2010/2011	2	MA4251/ST4238	Stochastic Processes II
2010/2011	1	MA3238/ST3236	Stochastic Processes I
2009/2010	2	MA4251/ST4238	Stochastic Processes II
2009/2010	1	MA3238/ST3236	Stochastic Processes I